# Sample report: GSM4293895_CD8

AstroGenetics: the physics of methylation, read on one sample

Specimen sorted CD8 T cells · platform sequencing

**Research use only. Not a diagnostic test.** Each reading is printed with its commissioning status: *commissioned* means it passed a test sealed before its data was seen; a reading outside what an instrument was tested on is still shown, and says so. The full technical record (intake, calibration, tare, noise gate, safeguards, integrity, and how to run it yourself) is in the HTML report.

## At a glance

| Instrument | Question it answers | Reading | State | Status |
|---|---|---|---|---|
| **IAM-Deconvolver** | Which cells, in what shares | | | not run (no array input) |
| **Met-A** | How far the neutrophil's tagging level has moved | | | not run (no array input) |
| **C-Score** | Are the departures clustered or scattered | | | not run |
| **IAM-A** | Is each mark still held, against the sample's own reference | **A 0.9083** | Normal | commissioned |

## AstroGenetics: the physics behind each reading

One law runs under every reading here: a mark held against thermal noise costs energy, and the loss of that hold shows as entropy, in bits. Each reading is a ratio to its healthy reference, so healthy reads 1. Its gauge runs from 0, the floor (below it a cell can no longer hold its identity), to saturation, the most the scale can register (every identity site fully scrambled). Both ends are derived from physics, not fitted to data.

| Instrument | What was measured | The physics |
|---|---|---|
| **IAM-A** | Held error ε = 0.0440: of the qualifying molecules, the share of sites left unwritten between written neighbors. This sample's loose share L = 0.238 predicts $\varepsilon_{pred}(L) = 0.0485$ from the self-reference line; **$A = \varepsilon/\varepsilon_{pred} = 0.908$**. | In bits: H(ε) = 0.261 bits per site. Holding energy E/kT = ln((1−ε)/ε) = 3.08. **Gauge: 0 · healthy 1 · saturation 10.31.** Healthy 1 is this sample's own predicted held error, 0.0485. Saturation: half the sites unwritten, nothing held (H = 1 bit per site). The floor sits at $A = 1.7\times10^{-8}$, effectively 0: the least error one ATP per site (M = 20.94) can buy against thermal noise. |

# IAM-A on its gauge

IAM-A · CD8 T cells, against its own reference · Normal

